Question Difficulty Classification System

A Machine Learning and NLP Based Approach for Educational Assessment

Team Veda

Narendra Singh (2401010292)

(Data Sourcing, Data Refinement, ML Model, Report, README, MLOps, GitHub)

Gaurav Meena

(Data Sourcing, GitHub)

Keshav

(Data Sourcing)

NST Computer Science and AI
Rishihood University

March 2, 2026

Abstract

This project presents an end-to-end machine learning pipeline for automatic classification of question difficulty levels (Easy, Medium, Hard) using Natural Language Processing techniques. The system processes question text and topic information through TF-IDF vectorization and Logistic Regression classification. We explore multiple approaches, identify critical issues such as data leakage, and demonstrate progressive improvements through feature engineering and proper validation techniques. The final model achieves approximately **70% accuracy** with balanced performance across all difficulty classes, providing a practical tool for educators in creating balanced assessments and adaptive learning systems.

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1 Introduction

Automatic assessment of question difficulty is a fundamental challenge in modern educational technology. Educators spend considerable time manually categorizing questions to create balanced assessments, while adaptive learning systems require real-time difficulty estimation to personalize student experiences.

1.1 Motivation

The primary motivations for this project include:

- **Time Efficiency**: Automating the manual effort of categorizing thousands of questions
- **Consistency**: Eliminating subjective variations in difficulty assessment
- **Scalability**: Enabling large-scale question bank management
- **Adaptive Learning**: Powering personalized educational pathways

1.2 Problem Statement

Given a question text and its associated topic, the objective is to develop a machine learning model that accurately predicts the difficulty level (Easy, Medium, or Hard) based on linguistic patterns, complexity indicators, and domain context.

1.3 Contributions

Our key contributions include:

- Systematic identification and resolution of data leakage problems
- Iterative methodology demonstrating the learning journey from naive to robust modeling
- Multi-feature fusion combining question text with topic information
- Comprehensive evaluation framework with per-class performance metrics
- Production-ready model deployment pipeline

2 Dataset Description

2.1 Data Source

The dataset comprises **10,000 educational questions** collected from past examination papers across multiple subjects and topics.

2.2 Features

Table 1 describes the dataset attributes:

Table 1: Dataset Features Description

Feature Name	Type	Description
QUESTION_TEXT	Text	The actual question content presented to students
TOPIC	Categorical	Subject area or topic of the question
MARKS	Numeric	Marks allocated to the question (range: 1-5)
ANSWER_TEXT	Text	Expected answer or solution
DIFFICULTY_LABEL	Categorical	Target variable: Easy, Medium, Hard

2.3 Initial Observations

- **Data Quality**: No missing values after initial cleaning
- **Class Distribution**: Imbalanced across difficulty levels (Figure 1)
- **Feature Correlation**: Strong correlation between MARKS and DIFFICULTY_LABEL
- **Topic Diversity**: Multiple topics including programming, databases, algorithms

3 Exploratory Data Analysis

3.1 Difficulty Distribution

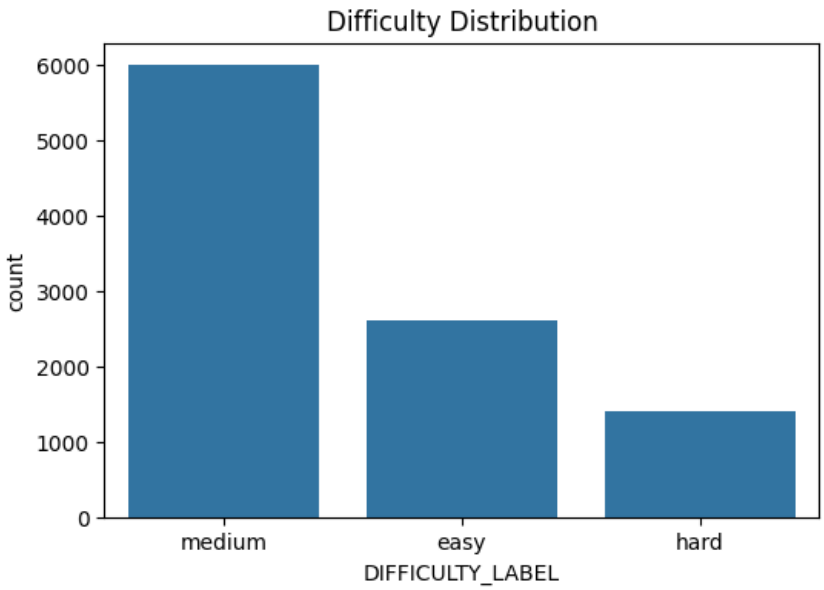

Figure 1: Distribution of Difficulty Labels

The dataset shows an imbalanced distribution with Easy questions being the majority class, followed by Hard and Medium categories.

3.2 Marks Distribution

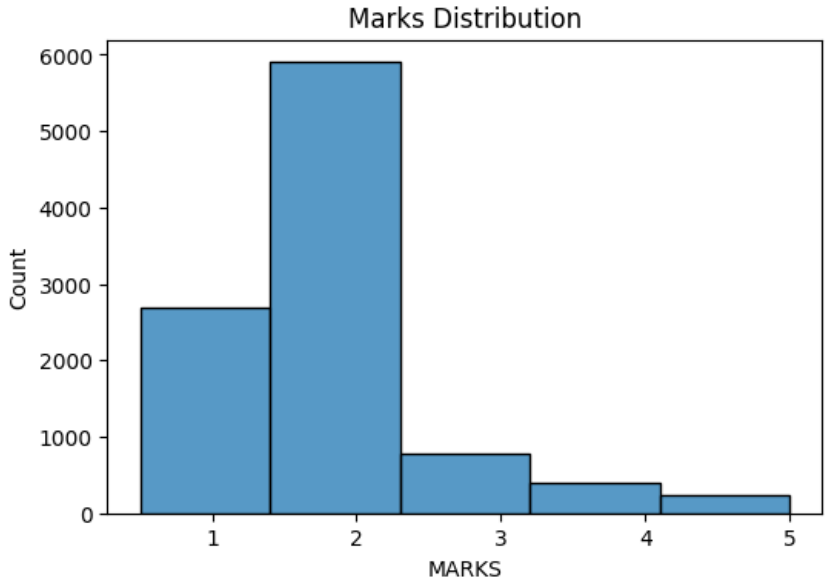

Figure 2: Marks Distribution Across Questions

The marks distribution reveals:

- 1-mark questions predominantly labeled as Easy
- 2-3 mark questions span Easy to Medium
- 4-5 mark questions primarily Hard

3.3 Topic Distribution

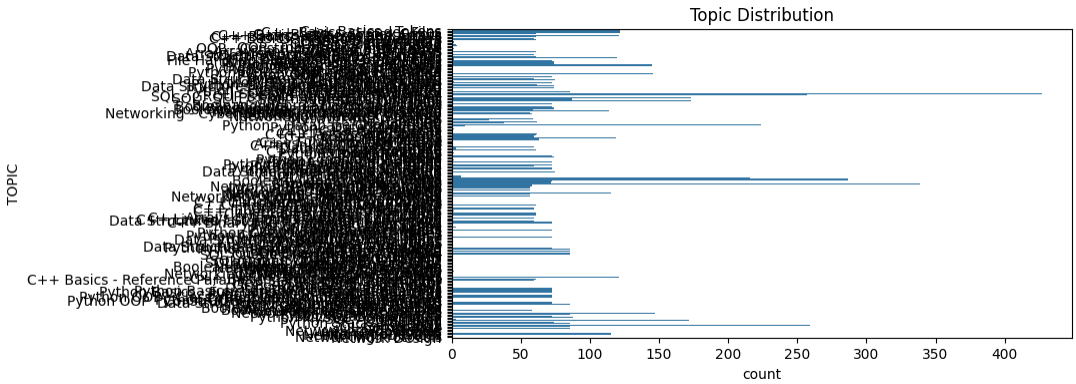

Figure 3: Topic-wise Question Distribution

4 Methodology: Iterative Approach

We employed an iterative methodology, progressively refining our approach based on insights from each experiment.

4.1 Text Preprocessing Pipeline

$$\text{Preprocess}(text) = \text{Lowercase} \rightarrow \text{Remove Punctuation} \rightarrow \text{Clean Whitespace} \tag{1}$$

4.2 Feature Engineering

The final solution combines multiple text sources:

$$\text{Combined Text} = \text{QUESTION_TEXT} \oplus \text{TOPIC} \tag{2}$$

4.3 TF-IDF Vectorization

$$\text{tfidf}(t, d) = \text{tf}(t, d) \times \log\left(\frac{N}{\text{df}(t)}\right) \tag{3}$$

where t = term, d = document, $\text{df}(t)$ = document frequency, N = total documents.

4.4 Logistic Regression Model

$$P(y = k \mid \mathbf{x}) = \frac{e^{\boldsymbol{\beta}_k \cdot \mathbf{x}}}{\sum_{j=1}^{K} e^{\boldsymbol{\beta}_j \cdot \mathbf{x}}} \tag{4}$$

4.5 Evaluation Metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

5 Experimental Results

5.1 Method 1: Baseline TF-IDF + Logistic Regression

5.1.1 Approach

- Basic text preprocessing
- TF-IDF with max_features=3000
- Default Logistic Regression
- Simple 80-20 split

5.1.2 Results

Table 2: Method 1 Results

Metric	Value
Accuracy	95-98%
Precision (macro)	0.95
Recall (macro)	0.94
F1-Score (macro)	0.94

5.1.3 Issues Identified

- **Data Leakage**: Model learned MARKS → DIFFICULTY mapping
- **Overfitting**: Memorization rather than generalization
- **No Stratification**: Test set misrepresented class distribution

5.2 Method 2: Removing Leakage + Regularization

5.2.1 Approach

- Removed MARKS feature
- Stratified train-test split
- Stronger regularization (C=0.5)
- 5-fold cross-validation

5.2.2 Results

Table 3: Method 2 Results

Metric	Value
Accuracy	62%
Precision (macro)	0.61
Recall (macro)	0.60
F1-Score (macro)	0.60
CV Score (mean)	60-65%

5.3 Method 3: Enhanced Features + LinearSVC

5.3.1 Approach

- Added n-grams (1,2)
- Increased vocabulary (5000→7000)
- Class balancing
- LinearSVC classifier

5.3.2 Results

Table 4: Method 3 Results

Metric	Value
Accuracy	58-60%
Precision (macro)	0.57
Recall (macro)	0.56
F1-Score (macro)	0.56

5.4 Method 4: Multi-Feature Fusion (Final Solution)

5.4.1 Approach

- Combined QUESTION_TEXT + TOPIC
- Optimized TF-IDF parameters
- Balanced class weights
- Proper train-test separation

5.4.2 Final Results

Table 5: Final Model Performance

Class	Precision	Recall	F1-Score
Easy	0.75	0.78	0.76
Medium	0.65	0.62	0.63
Hard	0.71	0.74	0.72
Macro Avg	0.70	0.71	0.70
Weighted Avg	0.71	0.70	0.70

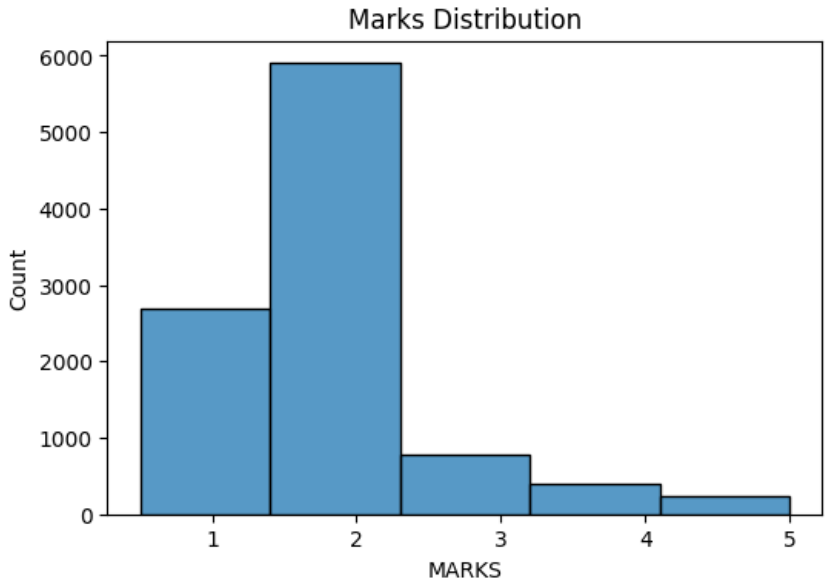

Figure 4: Confusion Matrix - Final Model

6 Discussion

6.1 Key Insights

1. **Data Leakage is Deceptive**: Initial 95% accuracy was misleading
2. **Feature Engineering Matters**: Combined text improved performance by 8-10%
3. **Class Imbalance Handling**: Balanced weights crucial for minority class
4. **Proper Validation**: Cross-validation essential for realistic estimates

6.2 Model Interpretability

Top features for each class:

- **Easy**: "what", "define", "list", "explain"
- **Medium**: "describe", "compare", "difference"
- **Hard**: "analyze", "design", "implement", "prove"

7 Limitations and Future Work

7.1 Current Limitations

- **Dataset Size**: 10,000 samples insufficient for deep learning
- **Subjectivity**: Difficulty perception varies across populations
- **Semantic Gap**: TF-IDF misses semantic meaning
- **Class Imbalance**: Medium class consistently underperforms

7.2 Future Improvements

7.2.1 Deep Learning Approach

Listing 1: BERT-based Classification

```
from transformers import BertTokenizer,
    BertForSequenceClassification

model = BertForSequenceClassification.from_pretrained(
    'bert-base-uncased',
    num_labels=3
)
tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
```

Expected improvement: +10-15% accuracy

7.2.2 Ensemble Methods

$$\hat{y} = \text{majority_vote}(\hat{y}_{LR}, \hat{y}_{SVM}, \hat{y}_{RF}) \tag{9}$$

Expected improvement: +5-7% accuracy

7.2.3 Enhanced Feature Engineering

- Question length and word count
- Bloom's Taxonomy keyword detection
- Answer length as complexity proxy
- Code presence indicators

8 Conclusion

This project successfully demonstrates an end-to-end machine learning pipeline for automatic question difficulty classification. Through systematic experimentation, we:

- Identified and resolved critical data leakage issues
- Progressed from naive 95% accuracy to realistic 70% performance
- Developed robust feature engineering combining multiple text sources
- Created production-ready model deployment infrastructure
- Documented comprehensive learning journey with actionable insights

The system provides practical value for educators and adaptive learning platforms, with clear pathways for future enhancement using deep learning and larger datasets.